

Problem 5

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Problem 1 For all x near 0, the inequalities $1 - \frac{x^2}{6} \leq \frac{\sin(x)}{x} \leq 1$ are true.

Use these inequalities to find $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$. **EXPLAIN.**

Explanation. $1 - x^2/6$ is a polynomial, so it is continuous at $x = 0$. That means $\lim_{x \rightarrow 0} \left(1 - \frac{x^2}{6}\right) = 1$. The function 1 is continuous since it is a constant, so $\lim_{x \rightarrow 0} 1 = 1$.

That means $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$ by the Squeeze Theorem, since $1 - \frac{x^2}{6} \leq \frac{\sin(x)}{x} \leq 1$ and $\lim_{x \rightarrow 0} \left(1 - \frac{x^2}{6}\right) = \lim_{x \rightarrow 0} 1 = 1$.